MINGXUAN LIU

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RESEARCH STATEMENT

My research vision is to develop generalizable AI systems for neuroscience and neuroimaging, spanning the full pipeline from acquisition to analysis to clinical translation. While my current work centers on fetal and perinatal brain MRI as one of the most data-scarce and clinically demanding domains in neuroimaging, I design methods with broad applicability to neuroscience at large. I am extending these foundations toward medical foundation models and agentic AI that can serve as trustworthy tools for both researchers and clinicians.

EDUCATION

09/2024-08/2028 Doctor of Philosophy (PhD), School of Biomedical Engineering, Tsinghua University
Advisor: Professor Qiyuan Tian

09/2021-08/2024 Bachelor of Engineering, School of Biomedical Engineering, Tsinghua University

09/2019-08/2021 Undergraduate Studies, Department of Chemical Engineering, Tsinghua University

09/2016-08/2019 High School Diploma, Changjun High School of Changsha, Changsha, Hunan

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AWARDS & HONORS

<u>*Research*</u>

A1. *Excellent Paper Competition Second Prize,* China Biomedical Engineering Conference & Medical Innovation Summit (BME), 2026
A2. *Travel Stipend Award,* International Society for Magnetic Resonance in Medicine (ISMRM), 2026, 2025, 2024
A3. *Travel Stipend Award,* ISMRM Workshop on Unlocking the Potential of Prenatal MRI, 2026 ($ 1,500)
A4. *Second Place in the Oral Session of AI for Image Reconstruction and Analysis,* Tsinghua CBIR Annual Retreat, 2026
A5. *Second Prize of Oral Presentation*, BEIHAI Summit (Competition judges: Charlotte Haug, Executive Editor of NEJM AI), 2025 (¥ 2,000)
A6. *First Place Winner of the autoPET IV Challenge,* MICCAI, 2025 (€ 1,800)
A7. *Second Place Winner of the VLM3D Challenge,* MICCAI, 2025
A8. *Merit Award (Top 1.6%),* Organization for Human Brain Mapping, 2025 ($ 850)
A9. *Invited Student Speaker Award,* The Second Graduate Academic Forum of the School of Biomedical Engineering and

the School of Clinical Medicine, Tsinghua University, 2025

A10. *Magna Cum Laude Merit Award (Top 15%),* ISMRM, 2024

A11. *Best Paper Award,* IEEE International Conference on Computer, Big Data, Artificial Intelligence, 2024

A12. *First Prize of 9th National Biomedical Engineering Innovation Design Competition for College Students (Top 5%)*, Chinese Society of Biomedical Engineering, 2024

A13. *Third Place Winner in the Poster Competition,* Singapore AI Health Summit (Competition judges: Rupa Sarkar, Editor-in-chief of The Lancet Digital Health; João Monteiro, Editor-in-chief of Nature Medicine; Pep Pàmies, Editor-in-chief of Nature Biomedical Engineering; and Prof Catherine R Lucey), 2023 ($ 3,000)

A14. *Scholarship for Science and Technology Innovation Excellence,* Tsinghua University, 2022

A15. *Second Prize of the Students Research Training Program (SRT),* Tsinghua University, 2022

<u>*Academic*</u>

A16. *National Scholarship (Top scholarship in China. 0.2% domestically)*, Ministry of Education, P.R. China, 2023 (¥ 8,000), 2024 (¥ 20,000), 2025 (¥ 20,000)

A17. *Outstanding Graduates of Beijing,* Beijing Municipal Commission of Education, 2024

A18. *Outstanding Graduate Award (Top 2%),* Tsinghua University, 2024

A19. *Scholarship for Academic Excellence*, Tsinghua University, 2020, 2022

RESEARCH GRANTS (Total ¥50 K)

SG1. Haoxiang Li, Mingxuan Liu. Automated Framework for Fetal Brain MRI Image Reconstruction and Disease Risk Prediction. <u>*Qi Yan Undergraduate Research Project*</u>, Beijing Municipal National Science Foundation, 2023. (¥50K)

ACADEMIC SERVICE

Journal Reviewer

Jr1. *Ad Hoc Reviewer*, Applied Intelligence

Jr2. *Ad Hoc Reviewer*, Biomedical Signal Processing and Control

Jr3. *Ad Hoc Reviewer*, Engineering Applications of Artificial Intelligence

Jr4. *Ad Hoc Reviewer*, IEEE Transactions on Cognitive and Developmental Systems

Jr5. *Ad Hoc Reviewer*, IEEE Transactions on Human-Machine Systems

Jr6. *Ad Hoc Reviewer*, Neural Networks

Jr7. *Ad Hoc Reviewer*, Neurocomputing

Jr8. *Ad Hoc Reviewer*, Pattern Recognition

Conference Reviewer

Cr1. *PC Member,* AAAI Conference on Artificial Intelligence (AAAI)

Cr2. *PC Member,* ACM Multimedia (MM)

Cr3. *PC Member,* ACM International Conference on Mobile Human-Computer Interaction (MobileHCI)

Cr4. *PC Member,* IEEE International Conference on Bioinformatics and Biomedicine (BIBM)

Cr5. *PC Member,* IEEE EMBS International Conference on Biomedical and Health Informatics (BHI)

Cr6. *PC Member,* British Machine Vision Conference (BMVC)

Cr7. *PC Member,* Conference on Health, Inference, and Learning (CHIL)
Cr8. *PC Member,* Conference on Neural Information Processing Systems (NeurIPS)
Cr9. *PC Member,* IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)
Cr10. *PC Member,* IEEE International Symposium on Biomedical Imaging (ISBI)
Cr11. *PC Member,* International Conference on Multimedia and Expo (ICME)
Cr12. *PC Member,* International Joint Conference on Neural Networks (IJCNN)
Cr13. *PC Member,* ISMRM & ISMRT Annual Meeting & Exhibition (ISMRM)
Cr14. *PC Member,* Medical Image Computing and Computer Assisted Intervention (MICCAI)
Cr15. *PC Member,* Medical Imaging with Deep Learning (MIDL)

International Conference Organization

IO1. *Co-organizer*, Medical Imaging Computing Seminar (MICS), 2026
IO2. *Co-organizer*, Beijing-Tsinghua Health AI Summit (BEIHAI Summit), 2025
IO3. *Co-organizer*, Annual OCSMRM Conference for Celebration of Excellence in Magnetic Resonance, 2024

Invited International Talks

T1. *Annotation-free deep learning for automated detection and segmentation of fetal germinal matrix-intraventricular hemorrhage in brain MRI*, China Biomedical Engineering Conference & Medical Innovation Summit (BME), 2026.
T2. *INSTA: Implicit Neural Spatio-Temporal Atlas from Thick-Slice Clinical Fetal Brain MRI*, ISMRM Workshop on Unlocking the Potential of Prenatal MRI in Washington, D.C., March, 2026
T3. *Maximizing Domain Generalization in Automatic Fetal Brain Biometry with Test-Time Adaptation*, The Second Graduate Academic Forum of the School of Biomedical Engineering and the School of Clinical Medicine, Tsinghua University, November, 2025
T4. *Brain in Utero Growth Among Fetuses with Germinal Matrix and Intraventricular Hemorrhage,* Beijing-Tsinghua Health AI Summit (To Rupa Sarkar, Editor-in-Chief of The Lancet Digital Health), October, 2025
T5. *FetalExtract-LLM: Structured Information Extraction from Free-Text Fetal MRI Reports Based on Privacy-Ensuring Open-weights Large Language Models,* MICCAI Workshop on Perinatal, Preterm and Paediatric Image Analysis (PIPPI) in Daejeon, South Korea, September, 2025
T6. *Detecting Fetal Germinal Matrix and Intraventricular Hemorrhage in Brain MRI Using Label-free Deep Learning,* Annual Meeting of Radiological Society of North America (RSNA) in Chicago, November, 2024
T7. *Image Quality Assessment using an Orientation Recognition Network for Fetal MRI (ORN-IQA),* ISMRM & ISMRT Annual Meeting & Exhibition (ISMRM) in Singapore, May, 2024
T8. *Label-free Image Quality Assessment of Fetal Brain MRI with Deep Learning (ORN-IQA),* Singapore AI Health Summit in Singapore, November, 2023
T9. *Skip-ST: Anomaly Detection for Medical Images Using Student-Teacher Network with Skip Connections,* IEEE International Symposium on Circuits and Systems (ISCAS), Online, May, 2023
T10. *Data Augmentation Using Image-to-image Translation for Tongue Coating Thickness Classification with Imbalanced Data,* IEEE Biomedical Circuits and Systems Conference (BioCAS), Online, October, 2022

Teaching Assistantship

- *Teaching Assistant*, Medical Image Processing (Undergraduate Course, 3 Units), School of Biomedical Engineering, Tsinghua University, 2024, 2025
- *Teaching Assistant*, Technological Innovation and Challenges (TIC) 3/4, Weixian College, Tsinghua University, 2024, 2025, 2026

RESEARCH ACHIEVEMENTS

Intellectual Property

P1. Tian Q, Li Z, Zheng J, Yang H, **Liu M**, Liao H. Method, Apparatus, and Electronic Device for Diffusion Prior Based Microstructure Modeling Optimization. (ZL 2025 1 0595484.7)

P2. Tian Q, Chen Y, Li Z, Zheng J, Yang H, **Liu M**. Method, Apparatus, and Electronic Device for Microstructure Parameter Estimation in Diffusion Magnetic Resonance Imaging. (Under Review)

P3. Tian Q, Bai X, **Liu M**. Method, Apparatus, and Device for Vision Language Large Model Construction for Chest Computed Tomography. (Under Review)

P4. Tian Q, **Liu M**, Qu H, Liao Y, Zhu J, Yang H, Yuan X. Method, Apparatus, and Electronic Device for Unsupervised Fetal Intracranial Hemorrhage Detection and Lesion Segmentation. (Under Review)

P5. Tian Q, Yang H, Qu H, **Liu M**, Li H, Liao Y, Zhu J, Jia F. Method, Apparatus, and Device for Three-Dimensional Reconstruction Based on Single-Thickness Slice Scanning of Fetal Brain Magnetic Resonance Imaging. (Under Review)

Journal Articles ([#]Co-first Author, [*]Corresponding Author)

<u>*First- and Corresponding-authored*</u>

J1. Y. Li[#], **M. Liu[#]**, X. Zhang[#], Y. Liao[#], W. Chen, K. Anmahapong, Z. Wang, Y. Chen, H. Yang, X. Bai, D. Yue, X. Liu, N. Sun, R. Hu, M. Kang, Y. Song, H. Lai, X. Zhou, J. Zhu, F. Jia, G. Ning, H. Qu[*], and Q. Tian[*]. Towards generalizable and expert-level fetal MRI report impression generation via adaptive fine-tuning of large language models. *npj Digital Medicine (NPJ DM)*. (Major Revision)

J2. Y. Hao[#], **M. Liu[#]**, J. Zhu[#], H. Yang, H. Li, G. Ning, Y. Liao, H. Qu[*], and Q. Tian[*]. PANDA: Patch-based unsupervised deep learning for brain anomaly detection via age prediction in fetal MRI. *Imaging Neuroscience (IMAG)*. 2026; 4: IMAG.a.1293.

J3. J. Zong[#], Y. Chen[#], **M. Liu[#]**, Y. Du, C. Wu, B. Wu, G. Zhou, and F. Qin[*]. MUIT-TTA: Annotation-Free Intracranial Hemorrhage Segmentation via Pseudo-Anomaly Synthesis and Test-Time Adaptation. *Pattern Recognition (PR)*, 2026; 180: 114143.

J4. **M. Liu[#]**, Y. Liao[#], H. Li, J. Zhu, H. Yang, Y. Hao, H. Qu[*], and Q. Tian[*]. Quality-label-free Fetal Brain MRI Quality Control Based on Image Orientation Recognition Uncertainty. *Medical Image Analysis (MIA)*, 2026; 110: 103994.

J5. **M. Liu[#]**, J. Tang[#], C. Yong, H. Li, J. Qi, S. Li, K. Wang, J. Gan, Y. Wang[*], and H. Chen[*]. Spiking-PhysFormer: Camera-Based Remote Photoplethysmography with Parallel Spike-driven Transformer. *Neural Networks (NN)*, 2025; 185: 107128.

J6. **M. Liu**, Y. Jiao, J. Lu, and H. Chen[*]. Anomaly Detection for Medical Images Using Teacher-Student Model with Skip Connections and Multi-scale Anomaly Consistency. *IEEE Transactions on Instrumentation and Measurement (IEEE TIM)*, 2024; 73: 1-15.

<u>*Co-authored*</u>

J7. H. Yang[#], Y. Liao[#], **M. Liu**, J. Zhu, H. Li, F. Jia, Z. Li, M. Zhang, J. Zheng, Z. Wang, J. Zhang, Y. Chen, Y. Li, L. Luo, J. Liu, H. Lai, X. Zhou, M. Kang, Y. Song, G. Ning, Z. Li, X. Zhang, H. Qu[*], and Q. Tian[*]. 3D fetal brain morphometry from single thick-slice MRI stacks using deep learning super-resolution. *Nature Biomedical Engineering (NBME)*. (Major Revision)

J8. M. Zhang[#], H. Yang[#], J. Xiao, W. Liu, T. Yin, Y. He, J. Zheng, Z. Li, **M. Liu**, W. Liu, C. Liao, B. Bilgic, J. R. Polimeni,

S. Y. Huang, Y. Liao, Z. Li[*], H. Qu[*], and Q. Tian[*]. SRNR: Deep learning-based MRI Super-Resolution using Noisy high-resolution Reference data. *Imaging Neuroscience (IMAG)*. (Major Revision)

J9. Y. Chen[#], Z. Li[#], Y. Wang, Y. Li, J. Zheng, H. Yang, **M. Liu**, T. Cukur, Z. Li, and Q. Tian[*]. MicroKAN: mapping human brain microstructure using diffusion MRI and convolutional Kolmogorov-Arnold Network. *NeuroImage (NIMG)*, 2026; 337: 122032.

J10. C. Wu[#], Y. Chen[#], Y. Du, J. Zong, J. Dong, **M. Liu**, Y. Peng, J. Fan, F. Qin[*], and C. Wang. Towards Practical Alzheimer's Disease Diagnosis: A Lightweight and Interpretable Spiking Neural Model. *Biomedical Signal Processing and Control (BSPC)*, 2026; 117: 109595.

J11. J. Zheng, Z. Li[*], W. Zhong, Z. Wang, Z. Li, H. Yang, **M. Liu**, X. Cao, C. Liao, D. H. Salat, S. Y. Huang, and Q. Tian[*]. Effects of diffusion MRI spatial resolution on human brain short-range association fiber reconstruction and structural connectivity estimation. *Imaging Neuroscience (IMAG)*, 2026; 4: IMAG.a.1089.

Conference Papers ([#]Co-first Author, [*]Corresponding Author)

First- and Corresponding-authored

C1. H. Li[#], **M. Liu[#]**, D. Varadarajan, Z. Hu, Q. Tian[*], and J. R. Polimeni. XSurfer: Reconstructing surface meshes of cerebral and cerebellar cortex from diverse MRI data using untrained neural networks. *European Conference on Computer Vision (ECCV)*, 2026. (Poster)

C2. Y. Li[#], **M. Liu[#]**, L. Chang[#], H. Yang, K. Anmahapong, Z. Wang, X. Hu, Y. Hao, H. Li, Y. Chen, X. Li, F. Jia, Y. Liao, H. Qu, and Q. Tian[*]. Maximizing Domain Generalization in Automated Fetal Brain Biometry. *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*. (Poster)

C3. X. Zhang[#], **M. Liu[#]**, Y. Chen, J. Zhu, K. Anmahapong, Y. Huang, Y. Zhang, H. Yang, Y. Liao, G. Ning, H. Qu, and Q. Tian[*]. ASTAR: Automated Induction of Standardized Medical Radiology Reporting Templates from Large-Scale Clinical Free-Text Corpora. *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*. (Poster)

C4. X. Hu[#], J. Huang[#], **M. Liu[#]**, K. Anmahapong, Y. Chen, Y. Luo, Y. Huang, X. Bai, Z. Li, Y. Liao, H. Qu, and Q. Tian[*]. FetalAgents: A Multi-Agent System for Fetal Ultrasound Image and Video Analysis. *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), 2026*. (Early Accept, Top 9%)

C5. **M. Liu[#]**, Y. Li[#], J. Zhu[#], H. Yang, Y. Huang, H. Li, Y. Chen, X. Bai, Y. Liao, H. Qu, and Q. Tian[*]. FetalExtract-LLM: Structured Information Extraction in Free-Text Fetal MRI Reports Based on Privacy-Ensuring Open-weights Large Language Models. *MICCAI Workshop on Perinatal, Preterm and Paediatric Image Analysis (PIPPI)*, 2025. (Oral)

C6. X. Bai[#], **M. Liu[#]**, Y. Chen, H. Yang, and Q. Tian[*]. Chest-OMDL: Organ-specific Multidisease Detection and Localization in Chest Computed Tomography using Weakly Supervised Deep Learning from Free-text Radiology Report. *Medical Imaging with Deep Learning (MIDL)*, 2025. (Poster, MICCAI VLM3D Challenge 2nd Place Winner)

C7. H. Li[#], **M. Liu[#]**, X. Bai, Y. Liao, J. Zheng, H. Yang, Z. Li, H. Qu, and Q. Tian[*]. FetalCSR: Multi-input Attention Fusion Network for Neural ODE-based Fetal Cortical Surface Reconstruction. *ICLR Workshop on AI for Children (AI4CHL)*, 2025. (Oral)

C8. X. Lin[#], **M. Liu[#]**, K. Liu, and H. Chen[*]. Spike-SLR: An Energy-efficient Parallel Spiking Transformer for Event-based Sign Language Recognition. *The British Machine Vision Conference (BMVC)*, 2024. (Poster)

C9. S. Li[#], **M. Liu[#]**, Y. Zhang, S. Chen, H. Li, H. Chen[*], and Z. Dou[*]. SAM-Deblur: Let Segment Anything Boost Image Deblurring. *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024. (Poster)

C10. **M. Liu[#]**, J. Gan[#], R. Wen[#], T. Li, Y. Chen, and H. Chen[*]. Spiking-Diffusion: Vector Quantized Discrete Diffusion Model with Spiking Neural Networks. *IEEE International Conference on Computer, Big Data and Artificial Intelligence (ICCBD+AI)*, 2024. (Best Paper Award)

C11. **M. Liu**, Y. Jiao, and H. Chen[*]. Skip-ST: Anomaly Detection for Medical Images Using Student-Teacher Network with Skip Connections. *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2023. (Oral)

C12. **M. Liu**, Y. Jiao, H. Gu, J. Lu, and H. Chen*. Data Augmentation Using Image-to-image Translation for Tongue Coating Thickness Classification with Imbalanced Data. *IEEE Biomedical Circuits and Systems Conference (BioCAS)*, 2022. (Oral)

Co-authored

C13. A. Wang#, Z. Li#, L. Zhang, X. Zhang, Y. Wang, X. Yang, **M. Liu**, X. Wang, H. Liao, and G. Ning*. GCM-Net: Anatomy-Aware Gaussian-Contrastive Multi-Layer Fusion Network for Abdominal Ultrasound Standard Plane Classification. *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2026. (Poster)

C14. Z. Gao#, C. Zhang#, **M. Liu**, Y. Chen, X. Hu, and Q. Tian*. View-Conditioned Collaborative Learning for Semi-Supervised Fetal Cardiac Ultrasound Analysis. *IEEE International Symposium on Biomedical Imaging (ISBI)*, 2026. (Oral)

C15. G. Zhou#, Y. Chen#, G. Ying, **M. Liu**, X. Bai, J. Zheng, B. Cui, Q. Tian*, and J. Lu*. WARPNet: Scale-wise Autoregressive Cross-modal Synthesis for Accurate and Detail-preserving MRI-to-PET Generation. *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, 2025. (Oral)

C16. Y. Wang#, Y. Chen#, S. Jiang, W. Yu, **M. Liu**, S. Zhu, F. Qin, and C. Wang*. DR-TTA: Dynamic and Robust Test-Time Adaptation Under Low-Quality MRI Conditions for Brain Tumor Segmentation. *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, 2025. (Oral)

C17. J. Huang, Y. Hao, Y. Luo, Z. Wang, **M. Liu**, Y. Chen, Y. Wang, L. Xiang, and Q. Tian*. autoPET IV challenge: Incorporating organ supervision and human guidance for lesion segmentation in PET/CT. *MICCAI autoPET IV challenge (MICCAI)*, 2025. (Oral, 1st Place Winner)

C18. J. Lu, **M. Liu**, and H. Chen*. PRTMTM: A Priori Regularization Method for Tooth-Marked Tongue Classification. *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2023. (Poster)

C19. S. Jiang, Y. Hong, C. Jiang, W. Chen, H. Chen, S. Zhu, B. Wu, **M. Liu**, Z. Zhu, F. Qin*, M. Tan, and Y. Chen*. GLeVE: Graph-Guided Lesion Grounding with Proposal Verification in 3D CT. *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2026. (Early Accept, Top 9%)

Conference Abstracts

First- and Corresponding-authored

C20. **M. Liu**, Y. Hao, H. Li, H. Yang, H. Qu, and Q. Tian*. Annotation-Free Deep Learning for Automated Detection and Segmentation of Fetal Germinal Matrix-Intraventricular Hemorrhage in Brain MRI. *China Biomedical Engineering Conference & Medical Innovation Summit (BME)*, 2026. (Oral, Excellent Paper Competition Second Prize)

C21. **M. Liu**, H. Li, H. Yang, Y. Liao, J. Zhu, J. Zheng, Y. Hao, Z. Li, Z. Li, Z. Wang, Y. Li, J. Huang, G. Ning, H. Qu, and Q. Tian*. Cortical Brain in Utero Growth Among Fetuses with Germinal Matrix and Intraventricular Hemorrhage (GMH-IVH). *ISMRM & ISMRT Annual Meeting & Exhibition (ISMRM)*, 2026. (Poster)

C22. **M. Liu**, Y. Hao, Y. Liao, H. Li, J. Zhu, H. Yang, Y. Chen, X. Bai, H. Qu, and Q. Tian*. Quality-Label-Free Stack-Level Quality Control Improves Fetal Brain Slice-to-Volume Reconstruction. *ISMRM Workshop on Unlocking the Potential of Prenatal MRI: Advances in Fetal Brain, Heart & Placenta Imaging*, 2026. (Poster)

C23. **M. Liu**, H. Li, Z. Li, H. Yang, J. Zheng, H. Qu, and Q. Tian*. Unsupervised Fetal Brain MRI Quality Assessment based on Orientation Prediction Uncertainty. *OHBM Annual Meeting (OHBM)*, 2025. (Poster, Merit Award, Top 1.6%)

C24. **M. Liu**, Y. Liao, J. Zhu, H. Li, H. Yang, J. Zheng, Z. Li, Z. Li, H. Qu, and Q. Tian*. FreeHemoSeg: Label-Free Deep Learning Framework for Automated Segmentation of Fetal Brain Germinal Matrix and Intraventricular Hemorrhage. *ISMRM & ISMRT Annual Meeting & Exhibition (ISMRM)*, 2025. (Poster)

C25. **M. Liu**, Y. Li, J. Zhu, H. Yang, Y. Huang, H. Li, Y. Chen, X. Bai, Y. Liao, H. Qu, and Q. Tian*. From Free Text to Usable Labels: Privacy-Ensuring Open-weights LLM-Enhanced Clinical Report Extraction for Fetal MRI. *Beijing-Tsinghua Health AI Summit (BEIHAI)*, 2025. (<u>Oral, Second Prize of Oral Presentation</u>)

C26. Y. Hao#, **M. Liu#**, H. Yang, H. Li, X. Bai, Y. Liao, H. Qu, and Q. Tian*. Comprehensive Evaluation of Unsupervised Image Enhancement for Volumetric Fetal Brain MRI. *Medical Imaging with Deep Learning (MIDL)*, 2025. (<u>Short Paper Track, Poster</u>)

C27. Y. Li#, **M. Liu#**, H. Yang, H. Li, X. Bai, Y. Liao, H. Qu, and Q. Tian*. Anatomy-guided Test-Time Adaptation for Automated Fetal Brain MRI Morphometry. *Medical Imaging with Deep Learning (MIDL)*, 2025. (<u>Short Paper Track, Poster</u>)

C28. H. Li#, **M. Liu#**, H. Yang, Y. Liao, H. Qu, J. R. Polimeni, and Q. Tian*. Self-Supervised Cortical Surface Reconstruction for Ultra High-resolution ex vivo 7T MRI. *Medical Imaging with Deep Learning (MIDL)*, 2025. (Poster)

C29. **M. Liu**, H. Li, J. Zhu, J. Zheng, H. Yang, Z. Li, Z. Li, and Q. Tian*. Detecting Fetal Germinal Matrix and Intraventricular Hemorrhage in Brain MRI Using Label-free Deep Learning. *Annual Meeting of Radiological Society of North America (RSNA)*, 2024. (<u>Oral</u>)

C30. **M. Liu**, H. Li, Z. Li, H. Yang, J. Zheng, X. Zhang, and Q. Tian*. Image Quality Assessment using an Orientation Recognition Network for Fetal MRI. *ISMRM & ISMRT Annual Meeting & Exhibition (ISMRM)*, 2024. (<u>Oral Power Pitch, Magna Cum Laude Merit Award, Top 15%</u>)

C31. **M. Liu**, H. Li, Y. Liao, H. Qu, and Q. Tian*. Label-free Image Quality Assessment of Fetal Brain MRI with Unsupervised Deep Learning. *Singapore AI Health Summit*, 2023. (<u>Poster Competition 3rd Place Winner, Top 3%</u>)

<u>Co-authored</u>

C32. X. Hu, **M. Liu**, H. Yang, Q. Tian*. INFANiTE: Implicit Neural Representation for High-resolution Fetal Brain Spatio-temporal Atlas Learning from Clinical Thick-slice MRI. *China Biomedical Engineering Conference & Medical Innovation Summit (BME)*, 2026. (<u>Poster</u>)

C33. Y. Li, **M. Liu**, H. Yang, and Q. Tian*. Maximizing Domain Generalization in Automated Fetal Brain Biometry. *China Biomedical Engineering Conference & Medical Innovation Summit (BME)*, 2026. (<u>Oral, Excellent Paper Competition Third Prize</u>)

C34. H. Yang, **M. Liu**, Y. Hao, Y. Liao, H. Qu, and Q. Tian*. Simple Baselines for Fetal Brain Unsupervised Anomaly Detection. *China Biomedical Engineering Conference & Medical Innovation Summit (BME)*, 2026. (<u>Oral, Excellent Paper Competition Second Prize</u>)

C35. X. Bai, **M. Liu**, M. Zhang, H. Yang, Z. Wang, Y. Luo, Y. Zhou, X. Han, and Q. Tian*. Preoperative CTA-based Deep Learning Model for Predicting AKI After TEVAR in Type B Aortic Dissection. *China Biomedical Engineering Conference & Medical Innovation Summit (BME)*, 2026. (<u>Oral, Excellent Paper Competition Third Prize</u>)

C36. Y. Zhang, Y. Xiang, and **M. Liu**. The Mask of Civility: Benchmarking Chinese Mock Politeness Comprehension in Large Language Models. *GloDAL International Conference in Applied Language Sciences (ALS)*, 2026. (<u>Oral</u>)

C37. H. Li, **M. Liu**, D. Varadarajan, Z. Hu, Q. Tian, and J. Polimeni*. Test-time optimization for cortical surface reconstruction across image resolutions/contrasts using untrained neural networks. *ISMRM & ISMRT Annual Meeting & Exhibition (ISMRM)*, 2026. (<u>Oral, Annual Meeting Planning Committee Selection Award</u>)

C38. H. Yang, **M. Liu**, Y. Liao, M. Zhang, J. Zhu, H. Li, F. Jia, Z. Li, Y. Li, J. Huang, Z. Wang, Z. Li, H. Qu, and Q. Tian*. Improving the Accuracy of Fetal Brain Age Prediction Using Neural Network Attribution Maps. *ISMRM & ISMRT Annual Meeting & Exhibition (ISMRM)*, 2026. (<u>Poster</u>)

C39. Y. Li, **M. Liu**, H. Yang, X. Hu, Y. Liao, K. Anmahapong, Z. Wang, J. Zhu, Y. Hao, Y. Chen, H. Li, Z. Li, F. Jia, H. Qu, and Q. Tian*. BioTTA: Maximizing Domain Generalization in Automatic Fetal Brain Biometry with Test-Time

Adaptation. *ISMRM & ISMRT Annual Meeting & Exhibition (ISMRM)*, 2026. (Oral, ISMRM Summa Cum Laude Merit Award, Top 5%)

C40. Y. Hao, **M. Liu**, J. Zhu, H. Yang, H. Li, J. Huang, Y. Liao, H. Qu, and Q. Tian*. Patch-based Unsupervised Deep Learning for Brain Anomaly Detection via Age Prediction in Fetal MRI. *ISMRM & ISMRT Annual Meeting & Exhibition (ISMRM)*, 2026. (Poster)

C41. J. Huang, K. Anmahapong, H. Li, **M. Liu**, J. Zhu, H. Yang, Y. Li, Y. Hao, F. Jia, Y. Liao, H. Qu, and Q. Tian*. Multi-Organ Volumetric Growth Trajectories in Fetuses with Hydronephrosis. *ISMRM & ISMRT Annual Meeting & Exhibition (ISMRM)*, 2026. (Poster)

C42. Y. Chen, G. Zhou, Y. Wang, **M. Liu**, X. Bai, J. Zheng, B. Cui, J. Lu, and Q. Tian*. AlignPET: Structure-Aligned MRI-to-PET Synthesis via Variational Autoregression Model for Ischemic Brain Lesions. *ISMRM & ISMRT Annual Meeting & Exhibition (ISMRM)*, 2026. (Oral Power Pitch)

C43. X. Hu, **M. Liu**, H. Yang, Y. Li, Y. Hao, H. Li, Y. Chen, Z. Li, J. Zhu, Y. Liao, G. Ning, H. Qu, and Q. Tian*. INSTA: Implicit Neural Spatio-Temporal Atlas from Thick-Slice Clinical Fetal Brain MRI. *ISMRM Workshop on Unlocking the Potential of Prenatal MRI: Advances in Fetal Brain, Heart & Placenta Imaging*, 2026. (Oral)

C44. X. Zhang, **M. Liu**, H. Yang, Y. Chen, J. Zhu, X. Bai, Y. Huang, Y. Hao, Z. Li, Y. Liao, G. Ning, H. Qu, and Q. Tian*. Automated Induction of Standardized Reporting Templates from Fetal Brain MRI Free-Text Corpora. *OHBM Annual Meeting (OHBM)*, 2026. (Poster)

C45. X. Hu, **M. Liu**, Y. Li, H. Yang, Y. Hao, H. Li, Y. Chen, Z. Li, J. Zhu, Y. Liao, G. Ning, H. Qu, and Q. Tian*. Benchmarking Spatiotemporal Fetal Brain Atlas Construction. *OHBM Annual Meeting (OHBM)*, 2026. (Poster)

C46. Y. Li, M. Liu, H. Yang, Y. Liao, K. Anmahapong, Z. Wang, J. Zhu, X. Hu, Y. Hao, Y. Chen, H. Li, Z. Li, F. Jia, H. Qu, and Q. Tian*. Towards Robustness Fetal Brain Development Monitoring and Universal Multi-Disease Diagnosis. OHBM Annual Meeting (OHBM), 2026. (Poster)

C47. Y. Chen, G. Zhou, Y. Wang, **M. Liu**, X. Bai, J. Zheng, B. Cui, J. Lu, and Q. Tian*. Modality-Agnostic PET Synthesis from Single-Modality Thick-Slice MRI via Structured MRI-PET Mapping. *OHBM Annual Meeting (OHBM)*, 2026. (Poster)

C48. Z. Li, J. Zheng, H. Yang, **M. Liu**, X. Hu, Y. Li, H. Liao, and Q. Tian*. DIMONER: Diffusion Model Optimization via Implicit Neural Representations with Signal Embedding. *OHBM Annual Meeting (OHBM)*, 2026. (Poster)

C49. Y. Wu, Y. Zhang, **M. Liu**, L. Zhang, and Q. Tian*. Prognostic AI Model for Myopia in Children & Adolescents Using Real-World Intervention Data: A 7-Center Study. *World Ophthalmology Congress (WOC)*, 2026. (Poster)

C50. Y. Wu, Y. Zhang, **M. Liu**, L. Zhang, and Q. Tian*. Artificial Intelligence-based Myopia Prediction in Children and Adolescents. *Asia-Pacific Academy of Ophthalmology Congress (APAO)*, 2026. (Poster)

C51. Y. Hao, **M. Liu**, J. Zhu, H. Yang, Y. Liao, H. Qu, and Q. Tian*. Unsupervised Anomaly Detection for Fetal Brain MRI using Two-Stage Denoising Autoencoder (ω-DAE). *OHBM Annual Meeting (OHBM)*, 2025. (Poster)

C52. H. Yang, **M. Liu**, Y. Liao, H. Li, J. Zhu, Z. Li, J. Zhang, J. Zheng, Z. Li, H. Qu, and Q. Tian*. FetalSR: Super-resolving High-isotropic-resolution Image Volume from Single Thick-slice Stack with Deep Learning for Fetal Brain Morphometry. *ISMRM & ISMRT Annual Meeting & Exhibition (ISMRM)*, 2025. (Oral, ISMRM Summa Cum Laude Merit Award, Top 5%)

C53. H. Li, **M. Liu**, Y. Liao, J. Zhu, J. Zheng, H. Yang, Z. Li, H. Qu, and Q. Tian*. CortexKAN: Multi-input KAN for fetal cortical surface reconstruction. *ISMRM & ISMRT Annual Meeting & Exhibition (ISMRM)*, 2025. (Poster)

C54. Y. Li, **M. Liu**, J. Zhu, H. Yang, J. Zheng, Z. Li, Y. Liao, H. Qu, and Q. Tian*. FetalSFUDA: Source-Free Unsupervised Domain Adaptation for Fetal Brain Extraction from Different Centers or MRI Sequences. *ISMRM & ISMRT Annual Meeting & Exhibition (ISMRM)*, 2025. (Poster)

C55. H. Li, Y. Liao, J. Zhu, **M. Liu**, J. Zheng, H. Yang, Z. Li, Z. Li, Q. Tian, and H. Qu*. Negative Impacts of Germinal Matrix-Ventricular Hemorrhage on Prenatal and Postnatal Brain Development. *ISMRM & ISMRT Annual Meeting &*

Exhibition (ISMRM), 2025. (Poster, ISMRM Summa Cum Laude Merit Award, Top 5%)

C56. Y. Chen, Z. Li, Y. Wang, Z. Li, J. Zheng, H. Yang, **M. Liu**, and Q. Tian*. DiffKAN: Convolutional Kolmogorov-Arnold Networks for Improved Diffusion MRI Microstructural Modeling. *ISMRM & ISMRT Annual Meeting & Exhibition (ISMRM)*, 2025. (Poster)

C57. J. Zhang, F. Lange, J. Andersson, J. Zheng, Y. Jing, H. Yang, **M. Liu**, Z. Li, W. Wu, Q. Tian, and Z. Li*. DeepEddy: high-quality fast eddy current and bulk motion correction using deep learning-based image synthesis and co-registration. *ISMRM & ISMRT Annual Meeting & Exhibition (ISMRM)*, 2025. (Oral, Magna Cum Laude Merit Award, Top 15%)

C58. Z. Li, J. Zheng, Z. Li, Z. Wang, **M. Liu**, G. Ning, H. Liao, and Q. Tian*. DIMOND++: Improving diffusion model optimization using diffusion priors. *ISMRM & ISMRT Annual Meeting & Exhibition (ISMRM)*, 2025. (Oral Power Pitch, Magna Cum Laude Merit Award, Top 15%)

C59. Y. Jing, Z. Li, Y. Yu, J. Zhou, Z. Li, J. Zheng, H. Yang, **M. Liu**, W. Wu, Q. Tian, and J. Lu*. High-fidelity Ultra-fast Diffusion Tensor Imaging in Stroke Patients Using Transfer Learning. *ISMRM & ISMRT Annual Meeting & Exhibition (ISMRM)*, 2025. (Poster)

C60. J. Zheng, C. Yang, W. Zhong, Z. Wang, H. Li, Z. Li, **M. Liu**, H. Yang, X. Cao, C. Liao, D. H. Salat, S. Y. Huang, and Q. Tian*. Impacts of diffusion MRI spatial resolution on the short-range association fiber reconstruction and connectivity estimation. *ISMRM & ISMRT Annual Meeting & Exhibition (ISMRM)*, 2025. (Poster)

C61. Y. Li, **M. Liu**, J. Zhu, H. Yang, J. Zheng, Z. Li, Y. Liao, H. Qu, and Q. Tian*. BrainSFUDA: Fetal Brain Extraction from Diffusion-weighted MR Images using Source-free Unsupervised Domain Adaption. *ISMRM Workshop on 40 Years of Diffusion*, 2025. (Oral Power Pitch)

C62. Y. Chen, Z. Li, S. Zhu, Z. Li, J. Zheng, H. Yang, **M. Liu**, and Q. Tian*. DiffKAN3D: Efficient and Accurate 3D Diffusion MRI Parameter Estimation for Real-Time Clinical Applications. *ISMRM Workshop on 40 Years of Diffusion*, 2025. (Poster)

C63. J. Zheng, C. Yang, W. Zhong, Z. Wang, H. Li, Z. Li, **M. Liu**, H. Yang, X. Cao, C. Liao, D. H. Salat, S. Y. Huang, and Q. Tian*. Effect of Diffusion MRI Acquisition and Nominal Spatial Resolutions on Fiber Reconstruction and Connectivity Estimation. *ISMRM Workshop on 40 Years of Diffusion*, 2025. (Oral)

C64. J. Zhang, F. Lange, J. Andersson, J. Zheng, Y. Jing, H. Yang, **M. Liu**, Z. Li, W. Wu, Q. Tian, and Z. Li*. DeepEddy: high-quality eddy current and bulk motion correction using deep learning-based image synthesis and co-registration. *ISMRM Workshop on 40 Years of Diffusion*, 2025. (Oral)

C65. Z. Li, J. Zheng, Z. Li, Z. Wang, **M. Liu**, H. Yang, G. Ning, H. Liao, and Q. Tian*. DIMOND++: Improving diffusion model optimization using diffusion priors. *ISMRM Workshop on 40 Years of Diffusion*, 2025. (Poster)

C66. Y. Zhang, **M. Liu**, Y. Wu, L. Zhang, C. Chai, Z. Liu, and Q. Tian*. Machine Learning and Deep Learning for Pediatric and Adolescent Myopia Prediction. *Beijing-Tsinghua Health AI Summit (BEIHAI)*, 2025. (Poster)

C67. Y. He, J. Zheng, H. Yang, Y. Chen, Z. Li, **M. Liu**, Q. Fan, Y. Liao, H. Qu, Q. Tian, J. Lu, and Z. Li*. HUMMID: High-fidelity Ultra-fast Macrostructure and Microstructure brain Imaging using Deep learning. *Annual Meeting of Radiological Society of North America (RSNA)*, 2025. (Poster)

C68. H. Li, **M. Liu**, J. Zheng, H. Yang, Z. Li, and Q. Tian*. FetalSurfer: Automated Fetal Cortical Surface Reconstruction. *ISMRM & ISMRT Annual Meeting & Exhibition (ISMRM)*, 2024. (Poster)